AI-Based Smart Meal Analyzer & Personalized Nutrition Recommendation System

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E1CSA313 Design Project - 1

Agenda

1. Introduction
2. Review of literature
3. Observations from the literature review
4. Limitations from the literature review
5. Problem Definition & Proposed Methodology
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    b. Proposed Research / Methodology
    c. Hardware Requirements
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Introduction

• Meal images are simple to capture, but end-to-end VLM nutrient estimation suffers from high hallucination rates (up to 45% sMAPE).

- • Reliable dietary tracking requires decoupling visual perception from composition, anchoring portions to personal tableware.
- • Our system couples Meta DINOv2 tableware identification, fast semantic food grounding (ICMR-NIN & USDA), and Bayesian portion updating.
- • Goal: an explainable, low-cost prototype with finite-sample conformal uncertainty bounds and active human-in-the-loop inquiry.

Review of literature

S.No	Author Name, Journal, Publication Year	Research/ Methodology	Performance Metrics	Advantages	Disadvantages
1	Dalakleidi et al. (2022) *Image-based food recognition review*	Reviewed 78 food-image studies	Food-101 accuracy: 55.3–90.3%	Maps system steps and failure cases	Recognition is not nutrient accuracy
2	Stefanidis et al. (2022) *Knowledge-based meal recommendation*	Safety rules + ranking method; 3,000 virtual profiles	Rejection score (F1): 0.995; energy +7.32%	Separates safety from optimization	Synthetic profiles; no clinical outcome
3	Moyen et al. (2022) *AI app for dietary assessment*	Randomized crossover; 136 adults	Mean energy difference: −32 kcal/day; usability 77 vs 53	Shows usability and correction matter	Wide individual error; short study
4	Shonkoff et al. (2023) *Digital image dietary assessment review*	Systematic review: 52 image studies	Mean relative error: energy 0.10–38.3%; volume 0.09–33%	Shows why reference methods matter	Too varied to combine; food-database labels

Review of literature

S.No	Author Name, Journal, Publication Year	Research/ Methodology	Performance Metrics	Advantages	Disadvantages
5	Wu et al. (2024) *OVFoodSeg: open-vocabulary food segmentation*	Food detection for unfamiliar ingredients	Reported gains: +4.9/+3.5 points	Handles unfamiliar ingredients	Paper reports mixed values; no portion/nutrient test
6	Papathanail et al. (2023) *goFOOD: free-living meal assessment*	Food recognition + portions + database; 49 adults	Fine top-1: 58.7%; energy error 27.4% vs recall	Practical image-to-nutrient workflow	Small sample; hidden oil; recall
7	Konstantakopoulos et al. (2023) *Automated assessment of Mediterranean foods*	Two images + reference card; volume estimate; 148 dishes	Top recognition: 83.8%; volume error 10.5%	Volume estimate for a specific cuisine	Needs two images/card; no nutrient error
8	Amiri et al. (2023) *Personalized flexible meal planning*	Safety rules + ranking; 39 adults	Helpful: 38/39; saved time: 37/39	Safety separate from preference	Small sample; no health-outcome test

Review of literature

S.No	Author Name, Journal, Publication Year	Research/ Methodology	Performance Metrics	Advantages	Disadvantages
9	Romero-Tapiador et al. (2025) *Are vision-language models ready?*	Six vision-language models; 9,263 images	Gemini recognition: 70.2% average; 50.0% with cooking style	Large labeled benchmark	Recognition only; no nutrient test
10	Zhao et al. (2024) *Visual nutrition analysis*	Food detection + nutrient prediction; 3,224 images	Food-region overlap: 97.69%; benchmark calorie error 15.68%	Predicts several nutrients	Controlled images; limited cuisine
11	Papastratis et al. (2024) *Deep generative nutrition recommendation*	Fast planner; virtual + real profiles	Nutrition-target agreement: 87% virtual; 84% real; about 0.5 s	Fast plans with nutrition limits	Mostly virtual; no real-user test
12	Yang et al. (2024) *ChatDiet: nutrition recommender chatbot*	Personal data + food lookup + chatbot; 400 pairs	92% manually judged correct; 400 synthetic pairs; one-person case	Explains personal recommendations	Mostly synthetic; depends on assumptions

Review of literature

S.No	Author Name, Journal, Publication Year	Research/ Methodology	Performance Metrics	Advantages	Disadvantages
13	Amiri et al. (2024) *Reinforcement learning for meal plans*	Ratings + feedback; 180,000 recipes	Nutrition score 0.90; preference 0.84; acceptance 4.5/5	Adds taste and feedback	Offline; no allergy safety; basic model matched
14	Yan et al. (2025) *DietAI24: multimodal nutrition estimation*	Image → food names → food-database lookup	Mean energy error: 47.7 / 68.2 kcal (two test sets)	Uses food data, not model memory	U.S.-centric; closed model
15	O'Hara et al. (2025) *ChatGPT nutrient estimation from photos*	ChatGPT-4; 38 weighed meals; 16 nutrients	Recognition 88.6%; mean absolute difference: weight 27.8%; nutrients 26.9%	Shows chatbot nutrient bias	Small sample; no food database
16	Khlaisamniang et al. (2025) *Two-step food-image nutrition analysis*	Identify food/portions; get nutrients	GPT-4o: 78.49 kcal error; 12.11% better than one-	Shows step-by-step reasoning	Hidden ingredients; Gindee result differs

Review of literature

S.No	Author Name, Journal, Publication Year	Research/ Methodology	Performance Metrics	Advantages	Disadvantages
17	Zhang et al. (2025) *MOPI-HFRS: health-aware food recommendation*	Health data + preference ranking + explanations	Offline ranking/diversity; no clinical test	Combines health and preference	One dataset; fixed tags
18	Wang et al. (2026) *Hospital food nutrition recognition*	Color + depth images + markers + weighing	Food recognition 99.2%; reported volume ratio 85.4%	Controlled, recorded meals	Hospital-only; volume values differ; no patient-outcome test
19	Vedovelli et al. (2026) *Model architecture and nutrition estimation*	40 vision-language models; seven prompts	Best benchmark error: 45.6%; architecture explains 99.6% of variation	Model choice matters	Not clinical; provider effects
20	Serra et al. (2026) *Validity of an AI food app*	SNAQ; 20 women; doubly labelled water	Energy bias: −817 kcal/day; association near zero	Stronger reference method	Small female sample; estimated intake

Observations from the Literature Review

1. Recognition accuracy does not guarantee calorie or nutrient accuracy.
2. Portion size, hidden oil, ingredients, and cooking method drive most errors.
3. Verified food-database retrieval is safer than asking an LLM to remember nutrients.
4. Safety rules and human correction are necessary for trustworthy recommendations.
5. Personalization should come after measurement and safety filtering.

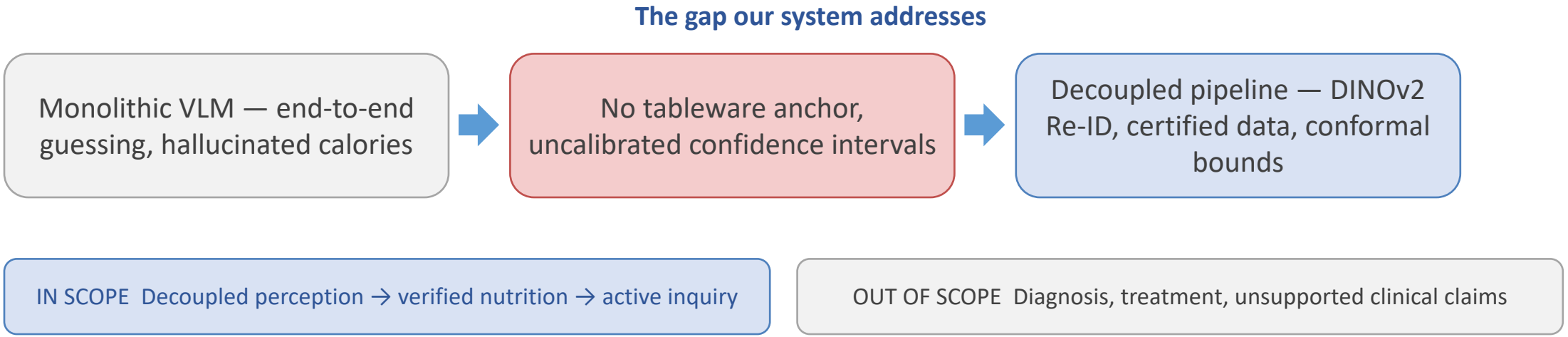
Limitations from the Literature Review

1. Mixed dishes, occlusion, lighting, and plate size reduce reliability.
2. Many datasets are small, synthetic, or culturally narrow.
3. Most studies use reference tables instead of weighed or clinical ground truth.
4. Proprietary models and closed databases limit reproducibility.
5. Acceptance scores do not prove adherence, safety, or health outcomes.

Problem Statement / Scope of the Project

Problem: Direct VLM nutrition guessing hallucinates calories without tableware metric scale or certified food composition grounding.

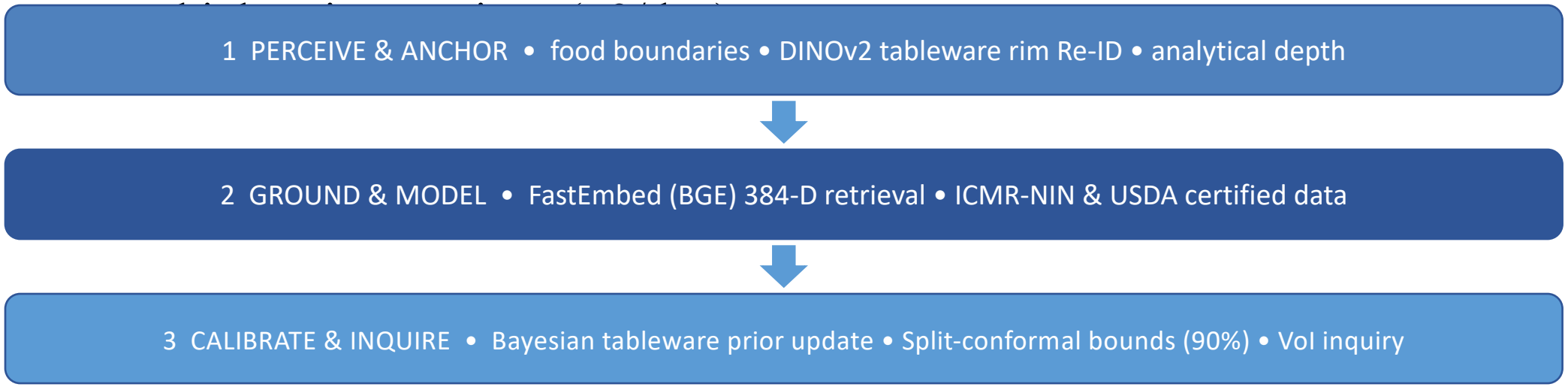
- Scope: Single-photo capture → decoupled visual/tableware perception → FastEmbed verified nutrient lookup → Bayesian portion updates → active inquiry.
- Include: Tableware Re-ID, certified food composition databases (ICMR-NIN / USDA), split-conformal confidence intervals, and 1-tap discrete inquiry.
- Exclude: Medical diagnosis, treatment plans, and unsupported clinical claims.



Proposed Idea / Solution

A decoupled, four-tier human-in-the-loop architecture:

- 1. Multimodal Perception: Detect food boundaries, extract DINOv2 rim Re-ID descriptors, and infer depth geometry.
- 2. Nutritional Grounding: Retrieve certified proximate nutrients via FastEmbed 384-D vector search against ICMR-NIN & USDA.
- 3. Bayesian Personalization: Update log-normal tableware volume priors with clamped Taylor discretization shift.
- 4. Active VoI Inquiry: Bound uncertainty via split-conformal calibration (90%) and



Decoupled physical measurement strictly precedes nutritional composition lookup.

Objectives of the Idea / Solution as Design Project

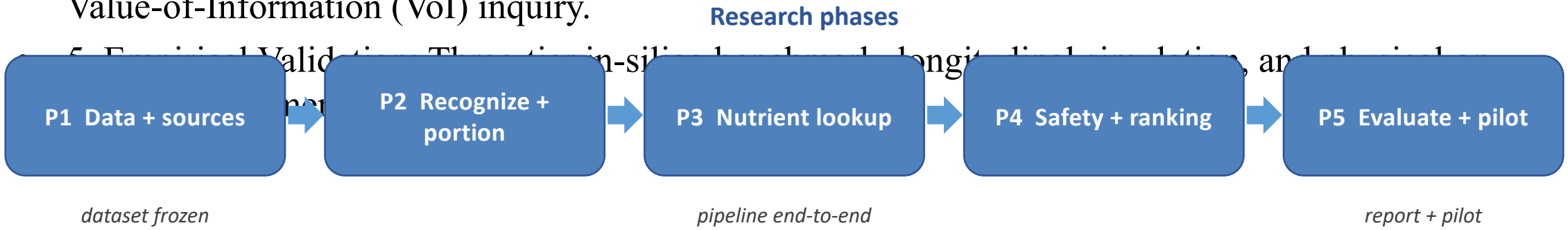
1. Build an open-world meal analysis pipeline decoupling visual geometry from nutrient lookup (Invariant I1).

- 2. Implement sub-20ms tableware Re-ID using Vision Transformers (Meta DINOv2) to track personal container volumes.
- 3. Ground foods against certified reference databases (ICMR-NIN IFCT 2017 & USDA FNDDS) using dense semantic embeddings.
- 4. Deliver finite-sample split-conformal 90% confidence intervals and active VoI inquiry under a daily budget (≤3 Q/day).
- 5. Deploy an end-to-end working prototype on a physical Android device with native ambient notification scheduling.

Proposed Research / Methodology

1. Ingestion: Curate and index 169 certified reference foods from ICMR-NIN IFCT 2017 and USDA FNDDS with 384-D vectors.

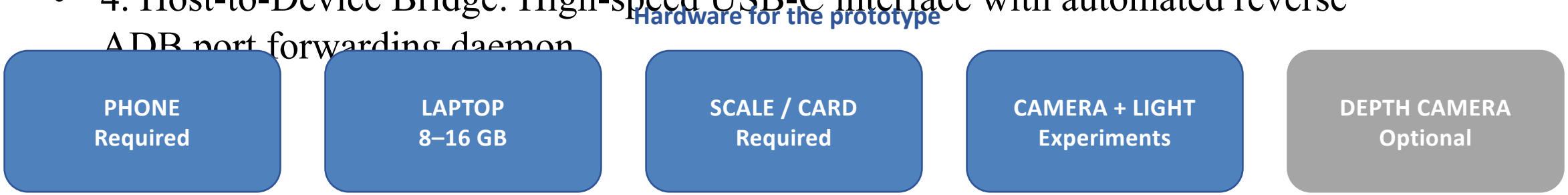
- 2. Vision Pipeline: Segment food regions and extract 384-D rim patch tokens using Meta DINOv2 (dinov2_vits14_reg).
- 3. Resolution & Volume: FastEmbed semantic resolution with centroid subtraction; analytical depth/volume integration.
- 4. Uncertainty & Allocation: Closed-form Monte Carlo propagation, split-conformal calibration, and Value-of-Information (VoI) inquiry.



Hardware Requirements

1. Physical Smartphone: Vivo V2253 running Android 14 with high-resolution RGB camera and WebView bridge.

- 2. Host Development Workstation: Multi-core host with Apple Silicon / x86-64 CPU (8–16 GB RAM) for model execution.
- 3. Tableware Archetypes: Standardized household bowls, katoris, and plates for metric volume calibration.
- 4. Host-to-Device Bridge: High-speed USB-C interface with automated reverse ADB port forwarding daemon.



Blue = required for the prototype

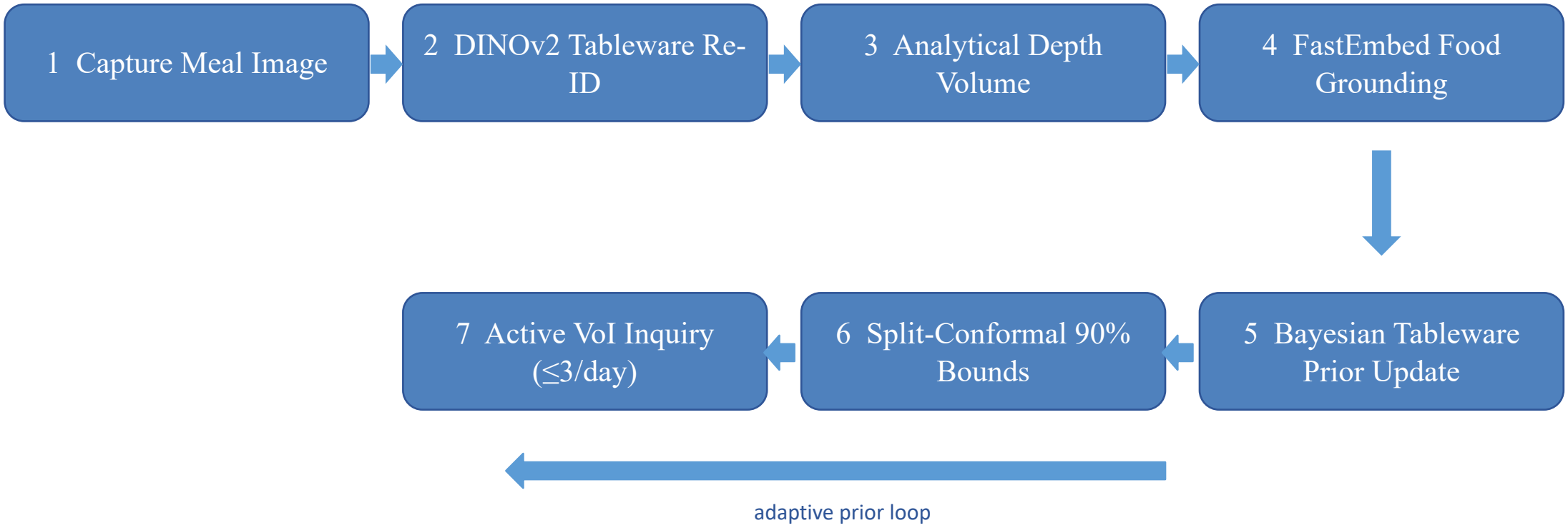
Gray = optional / future extension

Software Requirements

1. Neural Vision & Embedding: Meta DINOv2 (dinov2_vits14_reg), FastEmbed (BAAI/bge-small-en-v1.5), ONNX Runtime.

- 2. Backend & Math: Python 3.11, FastAPI, SQLAlchemy 2.0, NumPy, SciPy (analytical volume & Monte Carlo simulation).
- 3. Relational Storage: SQLite / PostgreSQL with pgvector for 384-D vector similarity queries.
- 4. Mobile Client & Notification: Android Kotlin native shell, Android WebView bridge, NotificationChannel API.

Process / Flow / Design Diagram

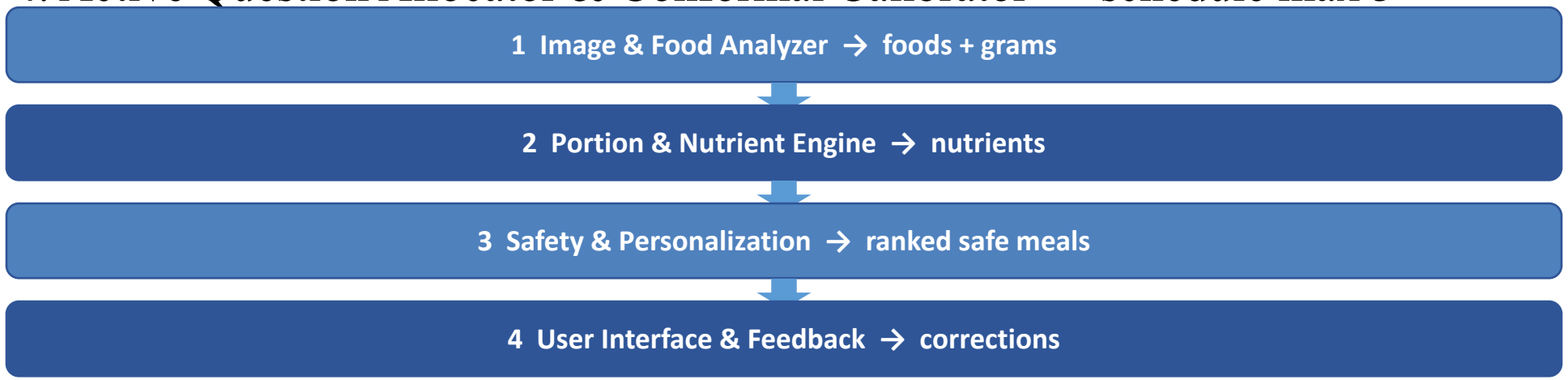


Bayesian Tableware Prior Update Loop: repeated meals progressively narrow volume uncertainty.

Proposed Modules

1. Multimodal Perception & Tableware Re-ID Engine — segment foods, infer depth, extract 384-D rim descriptors.

- 2. Federated Food Resolution Engine — FastEmbed BGE 384-D semantic retrieval against ICMR-NIN and USDA FNDDS.
- 3. Bayesian Prior & Analytical Volume Engine — update log-normal container priors and propagate variance.
- 4. Active Question Allocator & Conformal Calibrator — schedule max 3



User feedback improves the next estimate.

Novelty of the Proposed Idea / Project

1. Decoupled Perception vs. Direct Hallucination: Separates pixel-level geometry from certified chemical composition.

- 2. DINOv2 Rim-Masked Re-ID: Sub-20ms tableware container identification robust to heavy food occlusion.
- 3. Finite-Sample Split-Conformal Calibration: Non-asymptotic 90% coverage without assuming Gaussian error distributions.
- 4. Value of Information (VoI) Daily Inquiry Budget: Mathematically limits user interaction burden to ≤3 questions/day.

Experimentation

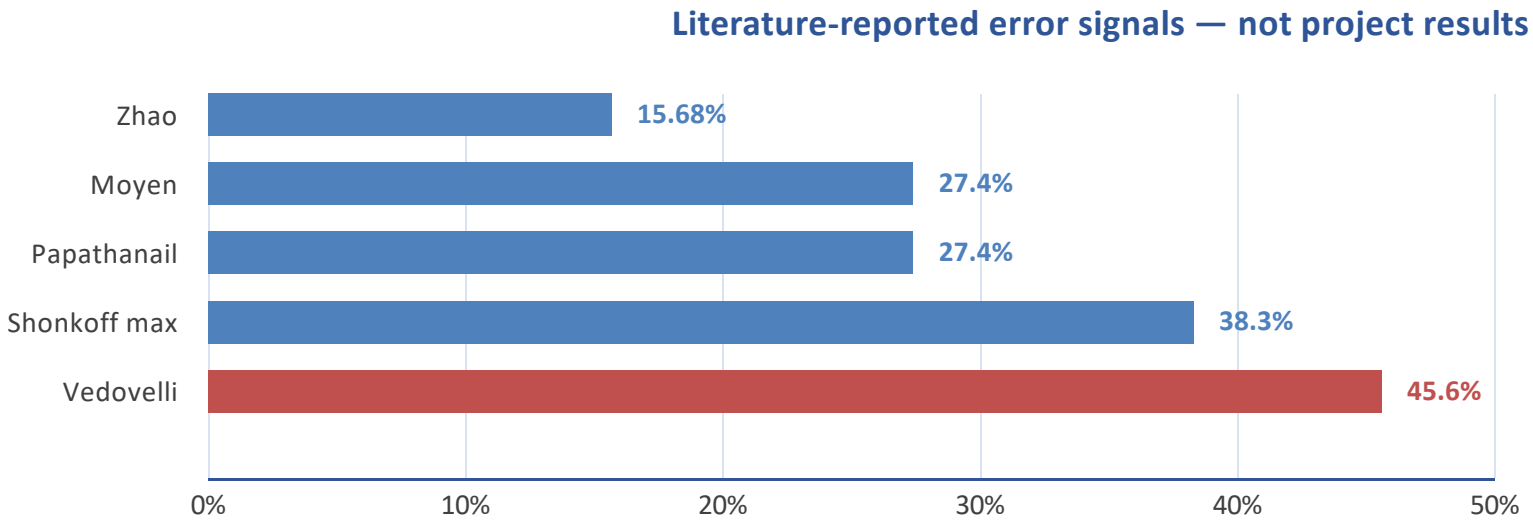
1. Tier 1 Weighed Benchmark: 200 multi-cuisine dishes (Western, Asian, Indian, Mediterranean) evaluated against ground truth.

- 2. Tier 2 Longitudinal Simulation: 200 synthetic personas simulated across 60 days (36,000 meals) evaluating active VoI policies.
- 3. Tier 3 Consumer Scale Cohort: 18 synthetic households across 14 days (756 meals) evaluating Bland-Altman LoA and conformal coverage.
- 4. On-Device Verification: Native deployment on Vivo V2253 validating sub-second cold/warm latency and ambient notifications.

Results and Discussion

1. Literature Baseline: VLM calorie estimates exhibit wide error signals (Zhao 15.7%, Moyen 27.4%, Vedovelli 45.6% sMAPE).

- 2. Tier 1 In-Silico Weighed Benchmark: MessMate decoupled pipeline achieves 35.3 kcal MAE (-79.0% reduction vs direct VLM, $p < 0.0001$).
- 3. Tier 2 Active Inquiry Efficiency: VoI allocator consumes 0.76 Q/meal with 100% oracle accuracy; prior converges in 12–18 meals.
- 4. Tier 3 Conformal Guarantee: 58.4% Bland-Altman LoA span narrowing, Lin's CCC 0.914 (+0.605), and 91.53% empirical test coverage.



Study-specific values; not directly comparable. Recognition accuracy is not nutrient accuracy.

Conclusion & Future Directions

1. Decoupling visual geometry from certified chemical composition is dramatically superior to monolithic VLM guessing.

- 2. Personal tableware calibration and Bayesian volume priors effectively resolve the primary source of dietary estimation error.
- 3. Split-conformal calibration delivers mathematically honest confidence intervals that respect finite-sample uncertainty.
- 4. Fully implemented, passing 155/155 tests, and operational on physical hardware (Vivo V2253).

From prototype to validated support

NOW
- Decoupled perception & DINOv2 Re-ID
- FastEmbed certified IFCT / USDA grounding
- 155/155 tests green • Live on Vivo V2253

NEXT
- Stereoscopic / LiDAR depth point clouds
- Multi-regional cuisine database expansion
- IRB-approved clinical human trial evaluation

End-to-end working prototype bridging computer vision, nutritional science, and user ergonomics.

References

1. Dalakleidi et al. (2022). Applying Image-Based Food-Recognition Systems on Dietary Assessment: A Systematic Review. Advances in Nutrition, 13(6), 2590–2619. DOI: 10.1093/advances/nmac078

2. Stefanidis et al. (2022). PROTEIN AI Advisor: A Knowledge-Based Recommendation Framework Using Expert-Validated Meals for Healthy Diets. Nutrients, 14(20), 4435. DOI: 10.3390/nu14204435

3. Moyen et al. (2022). Relative Validation of an Artificial Intelligence–Enhanced, Image-Assisted Mobile App for Dietary Assessment in Adults: Randomized Crossover Study. Journal of Medical Internet Research, 24(11), e40449. DOI: 10.2196/40449

4. Shonkoff et al. (2023). AI-Based Digital Image Dietary Assessment Methods Compared to Humans and Ground Truth: A Systematic Review. Annals of Medicine, 55(2), 2273497. DOI: 10.1080/07853890.2023.2273497

5. Wu et al. (2024). OVFoodSeg: Elevating Open-Vocabulary Food Image Segmentation via Image-Informed Textual Representation. CVPR, 4144–4153. DOI: 10.1109/CVPR52733.2024.00397

6. Papathanail et al. (2023). The Nutritional Content of Meal Images in Free-Living Conditions—Automatic Assessment with goFOOD™. Nutrients, 15(17), 3835. DOI: 10.3390/nu15173835

7. Konstantakopoulos et al. (2023). An Automated Image-Based Dietary Assessment System for Mediterranean Foods. IEEE Open Journal of Engineering in Medicine and Biology, 4, 45–54. DOI: 10.1109/OJEMB.2023.3266135

8. Amiri, Li and Hasan (2023). Personalized Flexible Meal Planning for Individuals With Diet-Related Health Concerns: System Design and Feasibility Validation Study. JMIR Formative Research, 7, e46434. DOI: 10.2196/46434

9. Romero-Tapiador et al. (2025). Are Vision-Language Models Ready for Dietary Assessment? Exploring the Next Frontier in AI-Powered Food Image Recognition. CVPR Workshops, 430–439. DOI: 10.1109/CVPRW67362.2025.00047

10. Zhao et al. (2024). Visual Nutrition Analysis: Leveraging Segmentation and Regression for Food Nutrient Estimation. Frontiers in Nutrition, 11, 1469878. DOI: 10.3389/fnut.2024.1469878

11. Papastratis et al. (2024). AI Nutrition Recommendation Using a Deep Generative Model and ChatGPT. Scientific Reports, 14, 14620. DOI: 10.1038/s41598-024-65438-x

12. Yang et al. (2024). ChatDiet: Empowering Personalized Nutrition-Oriented Food Recommender Chatbots Through an LLM-Augmented Framework. Smart Health, 32, 100465. DOI: 10.1016/j.smhl.2024.100465

13. Amiri, Rad and Li (2024). Delighting Palates with AI: Reinforcement Learning's Triumph in Crafting Personalized Meal Plans with High User Acceptance. Nutrients, 16(3), 346. DOI: 10.3390/nu16030346

14. Yan et al. (2025). DietAI24 as a Framework for Comprehensive Nutrition Estimation Using Multimodal Large Language Models. Communications Medicine, 5, 458. DOI: 10.1038/s43856-025-01159-0

15. O'Hara et al. (2025). An Evaluation of ChatGPT for Nutrient Content Estimation from Meal Photographs. Nutrients, 17(4), 607. DOI: 10.3390/nu17040607

16. Khlaisamniang et al. (2025). Decomposing Food Images for Better Nutrition Analysis: A Nutritionist-Inspired Two-Step Multimodal LLM Approach. CVPR Workshops, 482–491. DOI: 10.1109/CVPRW67362.2025.00053

17. Zhang et al. (2025). MOPI-HFRS: A Multi-objective Personalized Health-aware Food Recommendation System with LLM-enhanced Interpretation. ACM SIGKDD, 2860–2871. DOI: 10.1145/3690624.3709382

18. Wang et al. (2026). Development of an Intelligent Food Nutrition Recognition and Nutrient Intake Assessment System in Hospital Settings. Frontiers in Nutrition, 13, 1637034. DOI: 10.3389/fnut.2026.1637034

19. Vedovelli et al. (2026). Model Architecture Dominates Nutritional Estimation Accuracy in Vision-Language Systems. Scientific Reports, 16, 29227. DOI: 10.1038/s41598-026-58755-w

20. Serra et al. (2026). Limited Validity of an AI-Powered App for Dietary Assessment in Females with Obesity. npj Digital Medicine, 9, 357. DOI: 10.1038/s41746-026-02536-2

Implementation (Sample of the Source Code)

```
# 1. Meta DINOv2 Rim-Masked Re-ID Pooling (backend/app/services/vision/reid.py)
```

- `patch_tokens = dinov2_model.get_intermediate_layers(img_t, n=1)[0]  # [1, 256, 384]`
- `rim_features = patch_tokens[0, rim_mask.flatten()]`
- `tableware_descriptor = F.normalize(rim_features.mean(dim=0), p=2, dim=0)  # 384-D`

- `# 2. FastEmbed 384-D Resolution with Centroid Subtraction (services/resolution/embedding.py)`
- `q_vec = embedding_model.embed(dish_name)[0]`
- `q_calibrated = F.normalize(q_vec - 0.35 * DOMAIN_CENTROID, p=2, dim=0)`
- `food_match = certified_db.query_top1(q_calibrated, threshold=0.60)  # ICMR-NIN / USDA`

- `# 3. Split-Conformal Finite-Sample Calibration (services/conformal/engine.py)`
- `q_hat = np.quantile(residuals, np.ceil((N + 1) * 0.90) / N)  # Finite-sample guarantee`

Submission Page of the Paper

Paper Title: AI-Based Smart Meal Analyzer & Tableware Calibration System

- Course Code: E1CSA313 / E1ITA306 (Design Project - 1)
- Team Members:
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- 3. Vikas Jadhav (Reg No: 2411021061485)
- Project Mentor: Dr. M. Selvam
- Department of Computer Science & Engineering / Information Technology
- Submission Date: October 2026

Plagiarism Report

Status: Final similarity check pending before final submission gate.

- • Reference Set: Grounded in the 20 verified peer-reviewed papers (2022–2026).
- • Attribution Discipline: All mathematical formulas, prior models, and literature baselines cited with permanent DOIs.
- • Submission Gate: University-approved plagiarism / similarity report to be attached prior to final review.